

CS-1600

Host Online Specification

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Sysmex Corporation

Revision History

Version	Date	Description
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1. Scope of Application

This specification applies in the serial data communication (based on RS-232C) and the TCP/IP communication (Ethernet) between the CS-1600 and the host computer.

2. Overview

The following functions can be performed by means of the communication between the CS-1600 and the host computer:

1. Receives analysis orders from the host computer.
2. Sends analysis results to the host computer.
3. Sends reference information about samples to the host computer.

3. Serial Data Communication

The bit serial voltage type, which conforms to RS-232C interface, is used for the serial data communication. The CS-1600 can be connected to the host computer via the serial interface port. The specification complies with the Bit Serial Interface Standard Guidelines of The Japan Society for Clinical Laboratory Automation.

3.1 Hardware

3.1.1. Connector

- The connector to the host computer is located on the CS-1600
- A 9-pin D-SUB, female connector is used. (A male connector is used on the CS-1600.)

3.1.2. Connector Signals

Pin No.	Signal Name		Signal Direction
1	NC		
2	Receive Data	RxD	IN
3	Transmit Data	TxD	OUT
4	Data Terminal Ready	DTR	OUT
5	Signal Ground	SG	—
6	Data Set Ready	DSR	IN
7	Request to Send (Note*1)	RTS	OUT
8	Clear to Send (Note*1)	CTS	IN
9	NC		

(Note*1)

Whether to perform flow control for CTS/RTS can be selected according to the settings.

When “None” is selected for in the flow control, the signals of CTS/RTS are ignored.

3.1.3. Communication Format

The data is communicated in the asynchronous, half duplex mode. The following items can be defined as follows: (Asterisk mark “*” indicates the factory default.)

Baud Rate : 300, 2400, 4800, *9600, 19200 (bps)

Code : 7 bit, *8 bit

Stop Bit : *1 bit, 1.5 bit, 2 bit

Parity : *None, Even, Odd

Interval : 0, 1, *2, 5, 10 (sec)

3.1.4. Signal Level

Level	Data Signal	Control Signal
+3V or higher	Logic “0”, Start bit	ON
-3V or lower	Logic “1”, Stop bit	OFF

3.2. Software

3.2.1. Communication Format

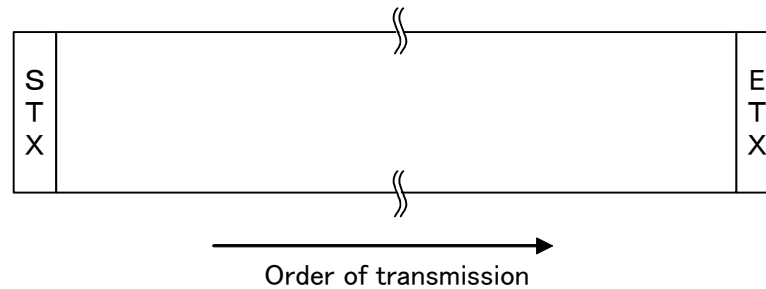
1) Code

ASCII codes are used for output.

2) Structure of Text

“STX” is sent prior to data and “ETX” is sent at the end of data.

One block should contain 255 characters or less. The analysis data is divided for transmission so that one block contains 255 or less characters.



3.2.2. Communication Protocol

The following 2 protocols are provided in the system.

(In the factory default, “Class B” is specified.)

- Class A:

One way transmission without requiring any response text from the host computer

- Class B:

Transmits data and then waits for the response text (ACK or NAK) from the host computer.

ACK: 06H, NAK: 15H

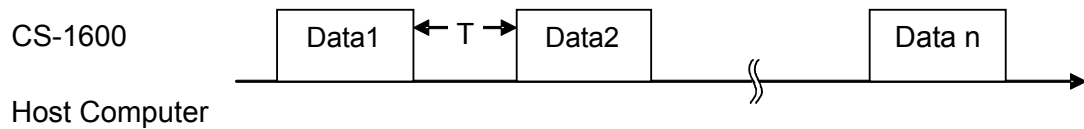
ACK and NAK can be sent between STX and ETX according to the settings.

The format of the response text can be defined in Response Text –Settings
Change-Serial Port - Host Computer- System Settings - Settings.

3.2.2.1 Outputting Analysis Result

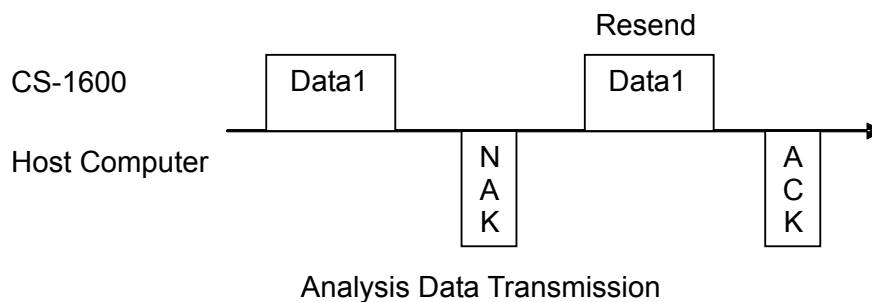
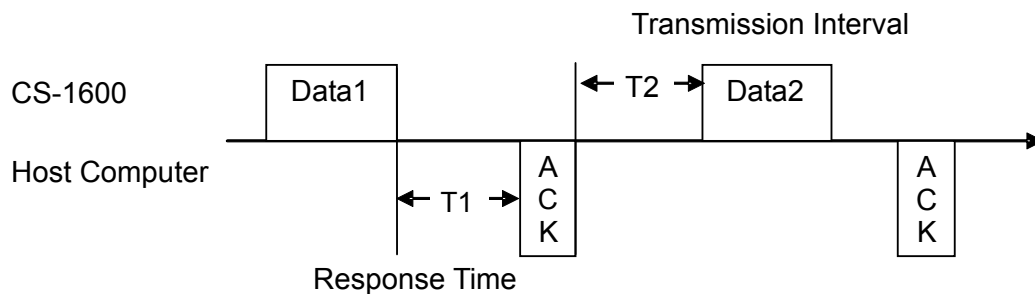
1) Communication Protocol (Class A)

The analysis result is output to the host computer taking the interval specified as T or longer interval. The interval can be selected from 0, 1, 2, 5, 10 sec according to the software settings.



2) Communication Protocol (Class B)

- (1) The CS-1600 sends an analysis result text.
- (2) The host computer sends ACK (06H) when successfully receiving the analysis result text or sends NAK (15H) when an error occurs.
- (3) The CS-1600 completes the transmission when receiving ACK (06H). When receiving NAK (15H), the CS-1600 resends the same text. However, the transmission is completed when receiving NAK after 3 retries.



Final Data (Final Report) and All Data

In this system, the automatic repeat analysis such as redilution analysis and reflex test can be performed when the analysis result exceeds the prescribed limit values. In that case, the results of first analysis and repeat analysis are integrated to produce the final data (final report). When the first analysis result does not exceed the limit values, it will be the final data.

However, when the repeat analysis is performed for all of the parameters in the first analyses, the repeat analysis results will be the final data.

Similarly, when the analysis is performed for the parameter whose replication setting is 2, the averaged result of the first and second analyses will be the final data. When the analysis whose replication setting is 1 is included the simultaneous analysis of multiple parameters, the result is included in the averaged result of the other parameters.

When another repeat analysis is performed for the averaged result, the data into which this repeat analysis result is integrated will be the final data (final report).

When the output target is Final Data, only the final data (final report) are output.

When the output target is All Data, it includes the analysis results before the final report.

The data output to the host computer from the CS-1600 per sample depends on the number of analysis parameters or analysis conditions. When the output target is set to "Final Data", the analysis results of all the parameters are output to the host computer at one time when all the analyses are completed. In the case of "All Data", the data transmission frequency per sample depends on the analysis conditions (replications and repeat analysis conditions).

Pattern 1

Analysis Condition	PT	APTT
Replication	1	1
Repeat analysis(dilution)	-	-
Reflect Analysis	-	-

Content of Analysis Data Format

Text Distinction Code II

			Setting for transmission	
PT	APTT	Text Distinction Code II	All Data	Final Data
●(First time)	●(First time)	"1"	★	★

Transmission Data1

F (Final)

Pattern 2(In the case of two replications)

Analysis Condition	PT	APTT
Replication	2	1
Repeat analysis(dilution)	-	-
Reflect Analysis	-	-

				Setting for transmission	
	PT	APTT	Text Distinction Code II	All Data	Final Data
Transmission Data1	●(First time)	●(First time)	"1"	★	
Transmission Data2	●(Second time)		"1"	★	
Transmission Data3	◎(Average)	◎(First time)	"2"	★	★

F (Final)

F (Final)

Pattern 3 (In the case of repeat analysis)

Analysis Condition	PT	APTT
Replication	1	1
Repeat analysis(dilution)	ON	-
Reflect Analysis	-	-

				Setting for transmission	
	PT	APTT	Text Distinction Code II	All Data	Final Data
Transmission Data1	●(First time)	●(First time)	"1"	★	
Transmission Data2	●(Repeat analysis)		"1"	★	
Transmission Data3	◎(Repeat analysis)	◎(First time)	"1 "	★	★
					F (Final)

F (Final)

Pattern 4 (In the case of repeat analysis on plural prrameters)

Analysis Condition	PT	APTT
Replication	1	1
Repeat analysis(dilution)	ON	ON
Reflect Analysis	-	-

				Setting for transmission	
	PT	APTT	Text Distinction Code II	All Data	Final Data
Transmission Data1	●(First time)	●(First time)	"1"	★	
Transmission Data2	●(Repeat analysis)	●(Repeat analysis)	"1"	★	★

F (Final)

F (Final)

Pattern 5 (In the case of repeat analysis 1 with two replications)

Analysis Condition	PT	APTT
Replication	2	1
Repeat analysis(dilution)	ON	-
Reflect Analysis	-	-

				Setting for transmission	
	PT	APTT	Text Distinction Code II	All Data	Final Data
Transmission Data1	●(First time)	●(First time)	"1"	★	
Transmission Data2	●(Second time)		"1"	★	
Transmission Data3	◎(Average)	◎(First time)	"2"	★	
Transmission Data4	●(Repeat analysis first time)		"1"	★	
Transmission Data5	●(Repeat analysis second time)		"1"	★	
Transmission Data6	◎(Repeat analysis Average)		"2"	★	
Transmission Data7	◎(Repeat analysis Average)	◎(First time)	"1"	★	★

F (Final)

F (Final)

Pattern 6 (In the case of repeat analysis 2 with two replications)

Analysis Condition	PT	APTT
Replication	2	1
Repeat analysis(dilution)	-	ON
Reflect Analysis	-	-

				Setting for transmission	
	PT	APTT	Text Distinction Code II	All Data	Final Data
Transmission Data1	●(First time)	●(First time)	"1"	★	
Transmission Data2	●(Second time)		"1"	★	
Transmission Data3	◎(Average)	◎(First time)	"2"	★	
Transmission Data4		●(Repeat analysis)	"1"	★	
Transmission Data5	◎(Average)	◎(Repeat analysis)	"1"	★	★

F (Final)

F (Final)

Pattern 7 (In the case of reflex analysis)

Analysis Condition	PT	APTT	Fbg
Replication	1	1	1
Repeat analysis(dilution)	-	-	-
Reflect Analysis	ON→Fbg	-	-

					Setting for transmission	
	PT	APTT	Fbg	Text Distinction Code II	All Data	Final Data
Transmission Data1	●(First time)	●(First time)		"1"	★	
Transmission Data2			●(Reflect Analysis)	"1"	★	
Transmission Data3	◎(First time)	◎(First time)	◎(Reflect Analysis)	"1"	★	★

F (Final)

The data marked with ★ is transmitted to the host computer.

● : Data which was actually analyzed

◎ : Data which was averaged or belong to one sample

3.2.2.2 Receiving Analysis Order

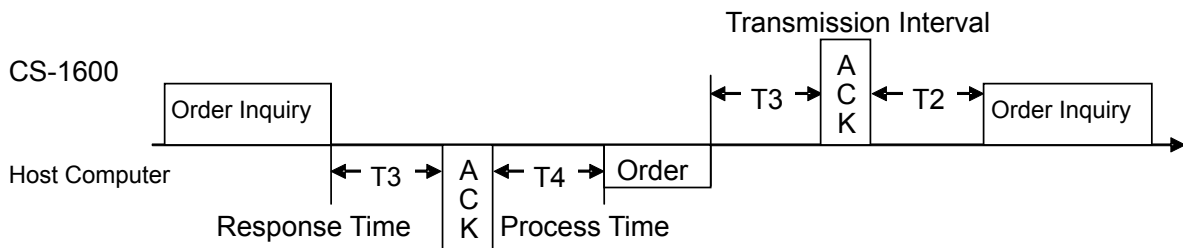
Receiving Analysis Order is performed by the following procedures;

- Sending Order inquiry from the IPU to the host computer

- (1) The CS-1600 sends an order inquiry text.
- (2) The host computer sends ACK (06H) when successfully receiving the order inquiry text or sends NAK (15H) when an error occurs.
- (3) The CS-1600 waits for the order information text when receiving ACK (06H).
When receiving NAK (15H), the CS-1600 resends the same text. However, the transmission is completed when receiving NAK after 3 retries.

- Sending order information from the host computer to the IPU

- (1) When receiving the order inquiry text, the host computer sends the order information text with an interval for 1 or more seconds after sending ACK (06H).
- (2) The CS-1600 sends ACK (06H) when successfully receiving the order information text. When an error occurs, the CS-1600 sends NAK (15H).
- (3) The host computer completes the communication when receiving ACK (06H).
When receiving NAK (15H), the host computer resends the same text.
However, more than three retries completes the transmission without sending NAK from the IPU to the host computer.



Order Inquiry and Order Information Texts Transmission

Transmission Interval:

The order information is output to the IPU taking the interval specified as T2 or longer interval. The interval can be selected from 0, 1, 2, 5, 10 sec according to the software settings.

Time-Out Period:

When the response time T1/T3 or the process time T4 times out, the transmission is completed. The time-out period should be 15 sec.

Process Time:

The process time, T4 must take an interval of 1 or more sec.

If no control signal line is used , T1 and T3 must take the interval of 0.2 sec or more.

(Note*2)

If no analysis parameter for the sample inquired by the CS-1600 is available in the host computer, set "000".

When there is no analysis parameter for the inquired sample and the later samples, set "999". The CS-1600 does not perform order inquiry for the samples in the subsequent tube positions after receiving "999".

3.2.3. Transmission Error

When a transmission error occurs, the transmission is interrupted and the error message is displayed in the CS-1600. Re-transmission is carried out by the operator.

Transmission errors occur in the following situations:

Error Code	Explanation	Action
40240	Frame error, parity error or overrun error is detected when a text is received. Interrupt the transmission	Check the host computer as well as the connector cable
40250	The control signal, DSR in the host computer is not activated. Interrupt the transmission	Check the host computer as well as the connector cable
40260	After the data is transmitted, no response is sent from the host computer within 15 seconds. Interrupt the transmission (Class B only)	Check the host computer as well as the connector cable
40270	After the data is transmitted, a response code other than ACK/NAK is sent from the host computer. Interrupt the transmission (Class B only)	Check the host computer as well as the connector cable
40280	When the data is transmitted, NAK is received 4 times. Interrupt the transmission (Class B only)	Check the host computer as well as the connector cable
40290	After the data is output, the control signal CTS is not turned on within 5 seconds. Interrupt the transmission (only for RTS/CTS control)	Check the host computer as well as the connector cable

Error Code	Explanation	Action
40300	1. STX is not received within 15 seconds after the text transmission is required. Interrupt the transmission (Class A) 2. STX is not received within 15 seconds after ACK when the text transmission is required. Interrupt the transmission (Class B) 3. STX for the subsequent subtext is not received within 15 seconds after the last subtext is received. Interrupt the transmission (Class A) 4. STX for the subsequent subtext is not received within 15 seconds after ACK is transmitted when the last subtext is received (Class B)	Check the host computer as well as the connector cable
40310	ETX is not received within 15 seconds after STX is received. Interrupt the transmission	Check the host computer as well as the connector cable
(Resend)	After the data is transmitted, NAK is sent from the host computer (Class B only)	Resend

4. TCP/IP Communication

4.1 Hardware

The following describes the network interface layer.

- Conforms to IEEE802.3.
- Use a communication cable of UTP category 5

4.2 Software

4.2.1 Data Link/Network/Transport Layer

- Conforms to TCP/IP protocol
- The IP address for the communication must be set by using the value defined in HC IP Address-Settings Change-TCP/IP-Host Computer-System Settings-Settings.
- The TCP port no. the communication must be set by using the value defined in Port No.-Settings Change-TCP/IP-Host Computer-System Settings-Settings.

4.2.2 Session Layer

The CS-1600, which is a client, is connected to the host computer, which is a server. The CS-1600 confirms the connection at the time of startup. When the connection fails, it is retried at regular intervals. If the server becomes down after the connection succeeded, the CS-1600 does not retry the connection.

4.2.3 Presentation Layer

“STX” (02H) is sent prior to data and “ETX” (03H) is sent at the end of data.

Note) In the presentation layer, an explicit response such as ACK and NAK is not given.

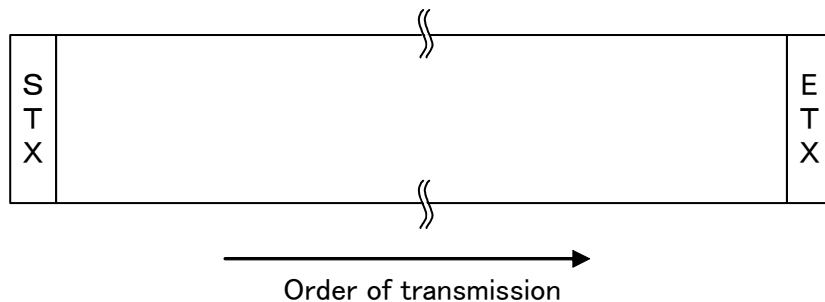


Figure 1 Text Format

4.2.4 Actions in the case of transmission error

When a transmission error occurs, the transmission is interrupted to display the error message in the CS-1600.

Re-transmission is carried out by the operator. Transmission errors occur in the following situations:

Table 1 Action for Transmission Error

Error Code	Explanation	Action
40340	Create error of the socket occurs or the connector cable is unplugged	Check the host computer as well as the connector cable
40360	Input and output errors occur in the physical layer due to disconnection of the HC communication path.	Check the host computer as well as the connector cable.

5. Text Format

The HC communications with the host computer are performed using the following text formats. They are common in the serial data and TCP/IP communications. The text formats of CA-500, CA-1500 format and CS-2000i format are described in this specification. Select the format corresponding to the host computer in the System Settings screen.

The CS-2000i format consists of the CA-1500 format and sample information text.

Note:

When CA-500 format is selected, up to 4-digit of Rack No. are displayed from the right. Though it doesn't apply for the Patient Name, space characters for 11-digit are reserved by space padding.

- **Analysis Data Text (Output)**

The analysis data is output.

There are two methods for output: Automatic transmission (real-time basis) after each analysis is completed.

- **Sample Information Text (Output) (Only for the CS-2000i format)**

The sample information checked by the analyzer before an analysis is output. The sample information text is output only when the CS-2000i format is selected as Format of Host Computer in the system settings. It is sent after the analysis data text of the final result is sent.

- **Order Inquiry Text (Output)**

The analysis parameters, or the sample No. and analysis parameters are inquired for prior to an analysis.

- **Order Information Text (Input)**

After transmitting the order inquiry text, order information text is received from the host computer for an order instruction.

* Class B communication is required to use the function of order inquiry/analysis order instruction by means of the serial-type communication.

5.1 Analysis Data Format

Parameter	Number of characters		
	CA-500	CA-1500 CA-2000i	
STX	1	1	(02H)
Text Distinction Code I	1	1	Fixed to "D"
Text Distinction Code II	1	1	"1", "2" The value is defined depending on the analysis condition.
Text Distinction Code III	2	2	Fixed to "21"
Block Number	2	2	"01" - "99"
Total Number of Blocks	2	2	"01" - "99"
Sample Distinction Code	1	1	"U", "E", "S", "C"
Date	6	6	(○○○○○○)
Time	4	4	(○○○○)
Rack Number	4	6	(○○○○○○)
Tube Position Number	2	2	(○○)
Sample ID Number	15	15	(○○...○○)
ID Information	1	1	"M", "A", "B"
Patient Name	11	15	(○○...○○)
Analysis Parameter, Data 1	9	9	
Analysis Parameter, Data 2	9	9	
	:	:	
Analysis Parameter Data N	9	9	
ETX	1	1	(03H)

Header Block

Data Block

- Order of Transmission

The order of transmission is from the top parameter to the bottom; the most significant digit first and the least significant digit last. Zero suppression is not performed.

- Text Distinction Code

A text distinction code is assigned to distinguish each text. There are three types of text distinction codes; from 1 to 3.

- Text Distinction Code 1 Fixed to "D"
- Text Distinction Code 2 1: Normal result data or final data.
2: Averaged result
- Text Distinction Code 3 Fixed to "21"

- Block Number, Total Number of Blocks

A text consists of one header block and 22 data block. When more than 22 data blocks exist, the text is divided for transmission so that it contains 255 or less characters.

The Block Number is the sequential number of the divided text. 01 through 99 is serially assigned.

The number of blocks is the sequential number of the divided text. 01 through 99 is serially assigned.

When the text is not divided, "01" is assigned to both.

- Sample Distinction Code

The types of analysis data are shown.

U: Routine analysis data

E: STAT analysis data

S: Calibration curve analysis data

C: Quality control analysis data

- Date and Time

The date and time when the analysis was performed are shown. The date format conforms to the system settings. Zero is appended for date and time by zero padding.

- Rack Number

Rack number is assigned to each sample rack. It is defined as shown below for expediency

Ordinary Sample : Consists of 6(4)-digit alphanumeric characters.

STAT Sample : Consists of 6(4)-digit alphanumeric characters.

Reagent Table : REAG00 is assigned. (When CA-500 format is used, REAG is assigned.)

Caution :

- When CA-500 format is selected, up to 4-digit of Rack No. are displayed from the right.

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of Rack Number between CS-1600 and each device.

- Tube Position Number

The tube position number indicates the sample tube position in the sample rack. Space characters are appended when a control is analyzed.

Ordinary Sample 01-10

STAT Sample 01-10

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of Tube Position Number between CS-1600 and each device.

- Sample ID Number

The sample ID number consists of 15 digit numeric characters. A hyphen "-"(2DH) may be inserted between numerals. The hyphen is included in the 15 digits. Zero suppression is not performed. When it consists of less than 15 digits, flush right is performed and blank digits are replaced with space characters (20H). When the Barcode Reading Error occurs, the sample No. beginning with "ERR00000000001" is sequentially assigned to the sample No.

- ID Information

The ID information indicates how the sample number is registered.

M: Manual entry

B: Read by barcode reader

A: Automatically assigned by the instrument

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of ID Information between CS-1600 and each device.

- Patient Name

The patient name consists of 15(11) characters with character codes including spaces but excluding control codes.

Caution :

- Though it doesn't apply for the Patient Name, space characters for 11-digit are reserved by space padding, when CA-500 format is selected,

- Analysis Parameter, Data

Parameter	Number of characters	
Parameter Code	3	(○○○)
Data	5	(○○○○○)
Flag	1	"□", "+", "—", "※", "!", "d", "<", ">"

The symbol "□" indicates a space character.

1) Parameter Code

The parameter code consists of two-digit host ID number defined in Basic Settings of Assay Group Definition and one-digit host ID number defined in Assay Parameter Settings of Assay Group Definition.

Refer to the parameter code list shown in the last part of this document.

The order of outputting Assay Group conforms to Measurement Group Settings of Settings. After parameters registered in Measurement Group1 are output in series, parameters registered in Measurement Group2 are output subsequently in series (but the parameters included in Measurement Group1 are excluded).

Formula parameters are output in the order of "Lupus Anticoagulant", "Protein C Global" and "Protein C Global / FV".

Example:

Following measurement group is output to the HC in the order of PT, APTT and Fbg.

Measurement group1		Measurement group2	
1	PT	1	PT
2	APTT	2	Fbg

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of the order of outputting between CS-1600 and each device.

2) Data

The analysis result consists of the digits (up to 5 digits) as defined in the assay parameter settings. The decimal point is not transmitted. The data as displayed in the Joblist is transmitted without the decimal point.

When the total number of the defined digits (integral part and decimal part) is less than 5 digits, space characters are inserted prior to the numeric value. When the analysis result consists of less than 5 digits, space characters are inserted prior to the numeric value. When the analysis result was not obtained due to analysis errors, the defined number of digits is masked with the mask characters. There are different mask characters depending on the error types. Explanation of each mask character is as follows;

"*": Analysis error

"/": Averaged result could not be calculated

+": Exceeding the number of display digits.

"-": Calculation parameter could not be calculated.

"X": No validated calibration curve is available for the calculation parameter.

NOTE) When the result exceeds the report limit, the limit value is output together with the flag as the result data.

Example)

Defined No. of digits	Normal analysis result		Analysis Result Error	Could not calculate averaged result	Exceeding the number of display digits	Could not calculate calculation parameter	No validated calibration curve is available for calculation parameter
4 digits for integral, 1 for decimal	□□□□1	99999	*****	/////	+++++	-----	XXXXX
2 digits for integral, 2 for decimal	□□□□1	□9999	□****	□////	□++++	□----	□XXXX
1 digit for integral, 2 for decimal	□□□□1	□□999	□□***	□□///	□□+++	□□---	□□XXX

The symbol "□" indicates a space character.

3) Flags

Space : No error

+ : Over the upper control limit

- : Under the lower control limit

< : Under the lower report limit

> : Over the upper report limit

* : Analysis error or difference in averaged data occurred

! : Coagulation time was obtained by re-dilution analysis

(Note*3)

d : Coagulation time was obtained by re-dilution analysis (Note*3)

(Note*3) :In CS-1600, the flag setting for the changing dilution ratio is selectable between "d" and "!" in the Other Service Settings-System Settings.

(Default setting: "d")

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of the flag of changing dilution ratio between CS-1600 and each device.

Caution!

In the case of CA-500 or CA-1500 format, there is a possibility that the uppermost digit is ignored and only lower 4 digits are received by the host computer. Moreover, as Ratio and INR results, upper 2 digits are ignored, and only lower 3 digits are received.

On CS-1600, when the digit number of result is over the digit number receivable by the host computer, it is impossible to receive the whole digits, and upper digits are not displayed on the host computer.

In the case of CA-500 or CA-1500 format, the set the digit number of assay parameter on the Assay Group Settings as follows. When a result is larger than the digit number, it is masked as +++++.

(The number of character '+', is depended on setup of digit number. When less than five characters, the figures are justified to the right and the space is padded on the left side.)

- Activity %, Concentration:

Set the digit number that an integral part and a decimal part make 4 figures in all.

- Ratio, INR:

When an integral digit and two decimal digits can be received by the host computer, set the digit number that 1 as integral part and 2 as decimal part. Otherwise, set the digit number that 2 as integral part and 2 as decimal part.

Ex.) In the case the Fbg C. is 1030.0mg/dL

Settings	wrong		correct	
	CS	HC	CS	HC
	Integral 4 Decimal 1	Integral 3 Decimal 1	Integral 3 Decimal 1	Integral 3 Decimal 1
display data on CS	1030.0		+++.	
sending data by CS	10300		□++++	
receiving data by HC		□0300		□++++
display data on HC		30.0		+++.

Ex.) In the case the PT INR is 12.34

Settings	wrong		correct	
	CS	HC	CS	HC
	Integral 3 Decimal 2	Integral 1 Decimal 2	Integral 1 Decimal 2	Integral 1 Decimal 2
display data on CS	12.34		+.++	
sending data by CS	□1234		□□+++	
receiving data by HC		□□234		□□+++
display data on HC		2.34		+.++

The symbol "□" indicates a space character.

5.2 Sample Information Text Format

* The sample information text is output only when the CS-2000i format is selected.

Parameter	Number of characters	
	CS-2000i	
STX	1	(02H)
Text Distinction Code I	1	Fixed to "D"
Text Distinction Code II	1	Fixed to "S"
Text Distinction Code III	2	Fixed to "21"
Block Number	2	"01" - "99"
Total Number of Blocks	2	"01" - "99"
Sample Distinction Code	1	"U", "E", "S", "C"
Date	6	(○○○○○○)
Time	4	(○○○○)
Rack Number	6	(○○○○○○)
Tube Position Number	2	(○○)
Sample ID Number	15	(○○...○○)
ID Information	1	"M", "A", "B"
Patient Name	15	(○○...○○)
H Flag	1	"□"
I Flag	1	"□"
L Flag	1	"□"
Sample Volume Flag	1	"V", "□", "-"
Reserved	86	Fixed space (20H)
ETX	1	(03H)

Header Block

Data Block

- Order of Transmission

The order of transmission is from the top parameter to the bottom; the most significant digit first and the least significant digit last. Zero suppression is not performed.

- Text Distinction Code

A text distinction code is assigned to distinguish each text. There are three types of text distinction codes; from I to III.

- Text Distinction Code I Fixed to "D"
- Text Distinction Code II Fixed to "S"
- Text Distinction Code III Fixed to "21"

- Block Number, Total Number of Blocks

A text consists of one header block and 22 data blocks. When more than 22 data blocks exist, the text is divided for transmission so that it contains 255 or less characters.

The Block Number is the sequential number of the divided text. 01 through 99 is serially assigned.
The number of blocks is the sequential number of the divided text. 01 through 99 is serially assigned.

When the text is not divided, "01" is assigned to both.

- Sample Distinction Code

The types of analysis data are shown.

U: Routine analysis data

E: STAT analysis data

S: Calibration curve analysis data

C: Quality control analysis data

- Date and Time

The date and time when the analysis was performed are shown. The date format conforms to the system settings. Zero is appended for the date and time by zero padding.

- Rack Number

The rack number is assigned to each sample rack. It is defined as shown below for expediency.

Ordinary Sample: Consists of 6-digit alphanumeric characters.

STAT Sample: Consists of 6-digit alphanumeric characters.

Reagent Table: REAG00 is assigned.

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of Rack Number between CS-1600 and each device.

- Tube Position Number

The tube position number indicates the sample tube position in the sample rack. Space characters are appended when a control is analyzed on the reagent table.

Ordinary Sample: 01-10

STAT Sample: 01-10

Reagent Table: □□ (The symbol "□" indicates a space character.)

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of Tube Position Number between CS-1600 and each device.

- Sample ID Number

The sample ID number consists of 15-digit numeric or alphabetical characters. A hyphen “-” (2DH) may be inserted between characters. The hyphen is included in the 15 digits. Zero suppression is not performed. When it consists of less than 15 digits, flush right is performed and blank digits are replaced with space characters (20H). When the Barcode Reading Error occurs, the sample number beginning with "ERR000000000001" is sequentially assigned to the sample number.

- ID Information

The ID information indicates how the sample number is registered.

M: Manual entry

B: Read by barcode reader

A: Automatically assigned by the instrument

- Patient Name

The patient name consists of 15 characters with character codes including spaces but excluding control codes.

- H Flag

The CS-1600 is fixed with space characters (20H)

- I Flag

The CS-1600 is fixed with space characters (20H)

- L Flag

The CS-1600 is fixed with space characters (20H)

- Sample Volume Flag

The result that the Sample Volume check gives is output. When the result is abnormal, "V" is output. When it is normal, space character (20H) is output. When it is not Sample Volume check, "-" is output. When the result is QC or Calibration, space character (20H) is output.

5.3 Order Inquiry Text Format

Parameter	Number of characters		
	CA-500	CA-1500 CS-2000i	
STX	1	1	(02H)
Text Distinction Code I	1	1	"R" (Fixed)
Text Distinction Code II (Order inquiry key)	1	1	"2" (Sample No.)
Text Distinction Code III	2	2	"21" (Fixed)
Block Number	2	2	"01" - "99"
Total Number of Blocks	2	2	"01" - "99"
Sample Distinction Code	1	1	Fixed space (20H)
Date	6	6	(○○○○○○)
Time	4	4	(○○○○)
Rack Number	4	6	(○○○○○○)
Tube Position Number	2	2	(○○)
Sample ID Number	15	15	(○○...○○)
ID Information	1	1	"M", "A", "B"
Reserved	11	15	All Space (20H)
Analysis Parameter, Data 1	9	9	
Analysis Parameter, Data 2	9	9	
:	:	:	
Analysis Parameter N, Data N	9	9	
ETX	1	1	(03H)

Header Block

Data Block

- Order of Transmission

The order of transmission is from the top parameter to the bottom; the most significant digit first and the least significant digit last. Zero suppression is not performed.

- Text Distinction Code

A text distinction code is assigned to distinguish each text. There are three types of text distinction codes; from 1 to 3.

- Text Distinction Code 1 "R" (Fixed)
- Text Distinction Code 2 (Order inquiry key) "2" : Sample No.
- Text Distinction Code 3 Fixed to "21"

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of Text Distinction Code 2 (Order inquiry key) between CS-1600 and each device.

- Block Number, Total Number of Blocks

A text consists of one header block and 22 data block. When more than 22 data blocks exist, the text is divided for transmission so that it contains 255 or less characters.

The block No. is the sequential number of the divided text. 01 through 99 is serially assigned.

The number of blocks is the sequential number of the divided text. 01 through 99 is serially assigned.

When the text is not divided, "01" is assigned to both.

- Date and Time

The date and time when the order inquiry text was transmitted are shown. The date format conforms to the system settings. Zero is appended for date and time by zero padding.

- Rack Number

Rack number is assigned to each sample rack. It is defined as shown below for expediency.

Ordinary Sample : Consists of 6(4)-digit alphanumeric characters.

STAT Sample : Consists of 6(4)-digit alphanumeric characters.

Caution :

- When CA-500 format is selected, up to 4-digit of Rack No. are displayed from the right.

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of Rack Number between CS-1600 and each device.

- Tube Position Number

The tube position number indicates the sample tube position in the sample rack.

Ordinary Sample 01-10

STAT Sample 01-10

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of Tube Position Number between CS-1600 and each device.

- Sample ID Number

The sample ID number consists of 15-digit alphanumeric characters. A hyphen "-"(2DH) may be inserted between numerals. The hyphen is included in the 15 digits. Zero suppression is not performed. When it consists of less than 15 digits, flush right is performed and blank digits are replaced with space characters (20H).

When the Barcode Reading Error occurs, the sample No. beginning with "ERR000000000001" is sequentially assigned to the sample No.

- ID Information

This flag indicates the method in which the sample ID number is registered.

- M : Manual entry
- A : Read by barcode reading error or sequential number automatically assigned by the instrument.
- B : B: Read by barcode reader

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of ID Information between CS-1600 and each device.

- Reserved

All spaces (20H)

Caution :

- When CA-500 format is selected, space characters for 11-digit are reserved by space padding,
- When CA-1500 format or CS-2000i format is selected, space characters for 15-digit are reserved by space padding,

- Analysis Parameter, Data

Parameter	Number of characters	
Parameter Code	3	(○○○)
Reserved	6	□□□□□□

The symbol "□" indicates a space character.

1) Parameter Code

The parameter code consists of the two-digit host ID Number defined in Basic Settings of Assay Group Definition and 0. The order inquiry can be performed for the analysis parameters that are defined as "Valid" in the Assay Group Settings and defined in the Measurement Group. Refer to the parameter code list shown in the last part of this document.

- * New analysis parameters may be added in the future. Please prepare a receive program that ignores the data when receiving the codes other than the above-mentioned codes.

2) Reserved

All spaces (20H)

5.4 Order Information Text Format

The same format as the Inquiry Text Format is used for the order information text format.

Parameter	Number of characters		
	CA-500	CA-1500 CS-2000i	
STX	1	1	(02H)
Text Distinction Code I	1	1	"S" (Fixed)
Text Distinction Code II (Order inquiry key)	1	1	"2" (Sample No.)
Text Distinction Code III	2	2	"21" (Fixed)
Block Number	2	2	"01" - "99"
Total Number of Blocks	2	2	"01" - "99"
Sample Distinction Code	1	1	"U", "E", "C"
Date	6	6	(○○○○○○)
Time	4	4	(○○○○)
Rack Number	4	6	(○○○○○○)
Tube Position Number	2	2	(○○)
Sample ID Number	15	15	(○○...○○)
ID Information	1	1	"M", "A", "B"
Patient Name	11	15	(○○...○○)
Analysis Parameter, Data 1	9	9	
Analysis Parameter, Data 2	9	9	
:	:	:	
Analysis Parameter N, Data N	9	9	
ETX	1	1	(03H)

Header Block

Data Block

- Order of Transmission

The order of transmission is from the top parameter to the bottom; the most significant digit first and the least significant digit last. Zero suppression is not performed.

- Text Distinction Code

A text distinction code is assigned to distinguish each text. There are three types of text distinction codes; from 1 to 3.

- Text Distinction Code 1 "S" (Fixed)
- Text Distinction Code 2 (Order inquiry key) "2"(Fixed)
- Text Distinction Code 3 Fixed to "21"

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of Text Distinction Code 2 (Order inquiry key) between CS-1600 and each device.

- Block Number, Total Number of Blocks

A text consists of one header block and 22 data blocks. When more than 22 data blocks exist, the text is divided for transmission so that it contains 255 or less characters.

The block Number is the sequential number of the divided text. 01 through 99 is serially assigned.

The number of blocks is the total number of the divided text. 01 through 99 is serially assigned. When the text is not divided, "01" is assigned to both.

- Sample Distinction Code

The types of analysis data are shown.

U : Routine analysis data

E : STAT analysis data

C : Quality control analysis data

- Date and Time

Sets the date and time when the inquiry was performed. The date format conforms to the system settings. Zero is appended for date and time by zero padding.

- Rack Number

Rack Number is assigned to each sample rack.

Ordinary Sample : Consists of 6(4)-digit alphanumeric characters.

STAT Sample : Consists of 6(4)-digit alphanumeric characters.

Reagent Table : REAG00 is assigned. (When CA-500 format is used, REAG is assigned.)

Caution :

- When CA-500 format is selected, up to 4-digit of Rack No. are displayed from the right.

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of Rack Number between CS-1600 and each device.

- Tube Position Number

The tube position number indicates the sample tube position in the sample rack. Space characters are appended when a control is analyzed on the reagent table.

Ordinary Sample 01-10

STAT Sample 01-10

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of Tube Position Number between CS-1600 and each device.

- Sample ID Number

The sample ID number consists of 15 digit numeric characters. A hyphen “-”(2DH) may be inserted between numerals. The hyphen is included in the 15 digits. Zero suppression is not performed. When it consists of less than 15 digits, flush right is performed and blank digits are replaced with space characters (20H).

- ID Information

This flag indicates the method in which the sample ID number is registered.

M : Manual entry
 A : Read by barcode reading error or sequential number assigned automatically
 B : Read by barcode reader

Caution :

Refer to [8. Difference List of Transmission Content] as the difference in the specification of ID Information between CS-1600 and each device.

- Patient Name

The patient name consists of 15(11) characters with character codes including spaces but excluding control codes.

Caution :

• When CA-500 format is selected, Send space characters for 11-digit,

- Analysis Parameter, Data

Parameter	Number of characters	
Parameter Code	3	(○○○)
Reserved	6	□□□□□□

The symbol "□"indicates a space character.

1) Parameter Code

The parameter code consists of the two-digit host ID Number defined in Basic Settings of Assay Group Definition and 0.

Refer to the parameter code list shown in the last part of this document.

The parameter codes “000” and “999” indicate abnormal situations.

000: There is no analysis parameter for the inquired sample.

999: There is no information on the inquired sample.

*1 If the analysis parameter for the inquired sample is not available in the host computer, set “000”. When the order information text includes the parameter code “000”, the sample will not be analyzed.

*2 If there is no analysis parameter for the inquired sample and later samples, set “999”

for the parameter code. When the CS-1600 receives the parameter code "999", the "No order is received from the host computer" error occurs. When the order information text includes the parameter code "999", the sample will not be analyzed. Furthermore, the order inquiries for the later samples in this batch are not performed. This action is prepared for the assumable case when there is no order information, and no sample No. or rack-tube position for the inquired sample. The order information text will be resend at the time of the analysis start for the following batch.

2) Reserved

All spaces (20H)

6. Parameter Code List

Assay Group	Parameter Code	Assay Group	Parameter Code
PT	04X	Reserved	45X
APTT	05X	Reserved	46X
Fbg	06X	Reserved	47X
TTO	08X	Reserved	50X
Hpt	09X	Reserved	51X
II	12X	Reserved	52X
V	15X	FDP	60X
VII	17X	DDimer	61X
VIII	18X	PFDP	62X
IX	19X	UFDP	63X
X	20X	PIC	64X
XI	21X	vWF-Ag medium	65X
XII	22X	vWF-Ag low	66X
Reserved	25X	Reserved	67X
Reserved	26X	Reserved	68X
Reserved	27X	Reserved	70X
Reserved	28X	Reserved	71X
Reserved	29X	Reserved	72X
AT3	30X	Reserved	80X
APL	31X	Reserved	81X
Plg	32X	Reserved	82X
PC (Chrom)	33X	Reserved	83X
Reserved	34X	Reserved	84X
Reserved	36X	Reserved	85X
Reserved	37X	Reserved	86X
Reserved	39X	Reserved	87X
Reserved	40X	Reserved	88X
Reserved	41X	Reserved	89X
Reserved	42X	Reserved	90X
Reserved	43X	Reserved	000
Reserved	44X	Reserved	999

000: There is no analysis parameter for the inquired sample.

999: There is no information on the inquired sample.

“X” is replaced with a one-digit host ID defined in the Assay Parameter Definition in when transmitting analysis results.

In the case of order inquiry, order information texts and the analysis data format of formula, “X” is replaced with 0.

- New analysis parameters may be added in the future. Please prepare a receive program that ignores the data when receiving the codes other than the codes shown in the parameter code list .

The following shows the default Host ID for assay parameter defined in the Assay Group Settings.

- 1: Time / dOD
- 2: Activity % / Concentration
- 3: Ratio
- 4: INR

- New analysis parameters may be added in the future. Please prepare a receive program that ignores the data when receiving the codes other than the codes shown above.

Caution :

- If the Host ID – Basic Settings - Assay Group Settings is changed, a different parameter is output. Please check the Host ID is correct.

7. Examples of Communication

Communication protocol is set as Class B in the format of CS-1600.

7.1 Inquiry of Analysis Order (CS-1600 => Host Computer)

When the order inquiry is performed from CS-1600

CS-1600	<STX>R2210101 090805093000000101 1234567890B 040 050 060 <ETX>
Host	<ACK>

7.2 Analysis Oder (Host Computer =>CS-1600)

When the host computer sends order to CS-1600

Host	<STX>S2210101U090805093000000101 1234567890B 040 <ETX>
CS-1600	<ACK>

7.3 Analysis Result (CS-1600=> Host Computer)

7.3.1 When the analysis is performed 1 time

CS-1600	<STX>D1210101U090805093000000101 1234567890B 041 174 042 505 044 111 <ETX>
Host	<ACK>
CS-1600	<STX>D1210101U090805093000000101 1234567890B V <ETX>
Host	<ACK>

7.3.2 When the analysis is performed 2 time

Analysis result (The first analysis)

CS-1600	<STX>D1210101U090805120000000203 1234567890B 041 204*042 401*044 144*051 962*<ETX>
Host	<ACK>

Analysis result (The second analysis)

CS-1600	<STX>D1210101U090805120000000203 1234567890B 041 206*042 399*044 143*<ETX>
Host	<ACK>

Analysis result (The final analysis)

CS-1600	<STX>D2210101U090805120000000203 1234567890B 041 205*042 400*044 144*051 962*<ETX>
Host	<ACK>
CS-1600	<STX>DS210101U090805120000000203 1234567890B V <ETX>

8. Difference List of Transmission Content

The CS-1600 supports formats for CA-500, CA-150 and CS-2000i. All of these formats are the same, however each device outputs different content by each device specification.

Please observe the precautions for use and ensure the operation of sending and receiving systems works with the host computer to use that format.

< CA-500 >

Contents	CA-500	CS-1600	Caution
Rack Number (Ordinary Sample)	4-digit numbers is output. The rack number appears in ascending order: '0001', '0002'... ('0099' of the following number is '0001'.)	4-digit <u>alpha-numeric</u> is output. Right 4-digit of the 6-digit rack numbers displayed in the CS-1600 is output.	Please check the host computer is able to receive alphabets. Please operate the CS-1600 to use numbers only (no alphabets) for the rack numbers if the host computer has trouble receiving alphabet characters.
Rack Number (STAT Sample)	4-digit numbers is output. The rack number(STAT Sample) appears in ascending order: '0100', '0200'... ('9900' of the following number is '0100'.)	4-digit <u>alpha-numeric</u> is output. Right 4-digit of the 6-digit rack numbers displayed in the CS-1600 is output.	Please check the host computer is able to receive alphabets. Please operate the CS-1600 to use numbers only (no alphabets) for the rack numbers if the host computer has trouble receiving alphabet characters.
Rack Number (Reagent Table)	No target exists. CA-500 does not operate if the control is set on the reagent table.	"REAG" is output.	Please check the host computer is able to receive alphabets.
Tube Position	STAT Sample : "00" Reagent table : No target exists. CA-500 does not operate if the control is set on the reagent table.	STAT Sample : 01~10 Reagent table : □□(□ is space.)	Please check the host computer is able to receive space.
ID Information	The CS-1600 does not use "C : Set by the host computer order information".		Please note the host computer does not output "C : Set by the host computer order information" and no other operation is working by using "C : Set by the host computer order information".

Order of outputting Assay Group	Output by fixing. Order is as follows PT, APTT, Fbg, TTO, HpT, II, V, VII, VIII, IX, X, XI, XII, AT3, APL, Plg, BCPC, +Fbg, -Fbg, SFDP, DD, PFDP, +DD, +SFD, +PFD	Output by <u>variable</u> . Order is as follows The order of outputting Assay Group conforms to Measurement Group Settings of Settings. After parameters registered in "Measurement Group1" are output in series, parameters registered in "Measurement Group2" are output subsequently in series	The CS-1600 does not output "+Fbg, -Fbg, +DD, +SFD, +PFD". The diluted assay group outputs the host ID of the assay group before dilution.
Flag setting for the changing dilution ratio	"!" is output.	"d" is output. *The flag setting for the changing dilution ratio is selectable between "d" and "!" in the Other Service Settings-System Settings.	Please set "!" as the flag in the Other Service Settings-System Settings.
Text Distinction Code 2 (Order inquiry key)	The CS-1600 does not use "1 : Rack No., Tube Position No."		Please note the host computer does not output "1 : Rack No., Tube Position No." and no other operation is working by using "1 : Rack No., Tube Position No."
How to inquiry order	The CS-1600 supports only "Smample ID No." for order inquiry. The CS-1600 does not support "Rack No., Tube Position No." "Smample ID No." : How to search analysis parameters by sample number on start of measurement. "Rack No., Tube Position No." : How to search sample number and analysis parameters by rack number and position		-

Caution :

Please ensure the number of digit on the Assay Parameter Settings - Assay Group Settings is set up properly. Refer to the caution of [Analysis Parameter, Data] in [5.1. Analysis Data Format] as the detail.

< CA-1500 >

Contents	CA-1500	CS-1600	Caution
Rack Number (Ordinary Sample)	6-digit numbers is output. The rack number appears in ascending order: '000001', '000002'... ('000099' of the following number is '000001'.)	6-digit <u>alpha-numeric</u> is output. The 6-digit rack numbers displayed in the CS-1600 is output.	Please check the host computer is able to receive alphabets. Please operate the CS-1600 to use numbers only (no alphabets) for the rack numbers if the host computer has trouble receiving alphabet characters.
Rack Number (STAT Sample)	"STAT H" is output.	<u>6-digit alpha-numeric</u> is output. The 6-digit rack numbers displayed in the CS-1600 is output.	Please check the host computer has received "STAT H" for the rack number of STAT sample and no other operation is working by using "STAT H" in the process. Please check the host computer is able to receive 6-digit alpha-numeric for Rack Number of STAT Sample.
Rack Number (Reagent Table)	"D1"~"D14" is output.	"REAG00" is output.	Please check the host computer has received "D1"~"D14" for the rack number of Reagent Table and no other operation is working by using "D1"~"D14" in the process. Please check the host computer is able to receive "REAG00" for Rack Number of Reagent Table.
Tube Position	STAT Sample : 01~05	STAT Sample : 01~10	-
ID Information	The CS-1600 does not use "C : Set by the host computer order information".		Please note the host computer does not output "C : Set by the host computer order information" and no other operation is working by using "C : Set by the host computer order information".
Order of outputting Assay Group	Output by <u>fixing</u> . Order is as follows PT, APTT, Fbg, TTO, NT, II, V, VII, X, VIII, IX, XI, XII, TT, AT3, Apl, Plg, PC, FDP, DD.PLUS,	Output by <u>variable</u> . Order is as follows The order of outputting Assay Group conforms to Measurement Group Settings of Settings. After parameters	-

	PSAc, LA1, LA2	registered in “Measurement Group1” are output in series, parameters registered in “Measurement Group2” are output subsequently in series	
Flag setting for the changing dilution ratio	“!” is output.	“d” is output. *The flag setting for the changing dilution ratio is selectable between "d" and "!" in the Other Service Settings-System Settings.	Please set “!” as the flag in the Other Service Settings-System Settings.
Text Distinction Code 2 (Order inquiry key)	The CS-1600 does not use “1 : Rack No., Tube Position No.”.		Please note the host computer does not output “1 : Rack No., Tube Position No.” and no other operation is working by using “1 : Rack No., Tube Position No.”.
How to inquiry order	The CS-1600 supports only “Smample ID No.” for order inquiry. The CS-1600 does not support “Rack No., Tube Position No.”. “Smample ID No.” : How to search analysis parameters by sample number on start of measurement. “Rack No., Tube Position No.” : How to search sample number and analysis parameters by rack number and position		-

Caution :

Please ensure the number of digit on the Assay Parameter Settings - Assay Group Settings is set up properly. Refer to the caution of [Analysis Parameter, Data] in [5.1. Analysis Data Format] as the detail.

< CS-2000i >

Contents	CS-2000i (CS-5100)	CS-1600	Caution
Rack Number (STAT Sample)	"STAT□□"(□ is space) is output.	6-digit <u>alpha-numeric</u> is output. The 6-digit rack numbers displayed in the CS-1600 is output.	Please check the host computer has received " STAT□□" for the rack number of STAT sample and no other operation is working by using " STAT□□" in the process. Please check the host computer is able to receive 6-digit alpha-numeric for Rack Number of STAT Sample.
Rack Number (LAS Sample)	"LXXXXX"(XXXXX is 5-digit numbers) is output. (Only CS-5100)	Not applicable	-
Tube Position	STAT Sample : 01~05 LAS Sample : 01 (Only CS-5100)	STAT Sample : 01~10 LAS Sample is not applicable.	-
Order of outputting Assay Group	Output by variable. Order is as follows The order of outputting Assay Group is an ascending order of HostID registered in Assay Group Setting	Output by variable. Order is as follows The order of outputting Assay Group conforms to Measurement Group Settings of Settings. After parameters registered in "Measurement Group1" are output in series, parameters registered in "Measurement Group2" are output subsequently in series	-
Flag setting for the changing dilution ratio	CS-2000i: "!" is output. CS-5100: "d" is output. *The flag setting for the changing dilution ratio is selectable between "d" and "!" in the System Settings.	"d" is output. *The flag setting for the changing dilution ratio is selectable between "d" and "!" in the Other Service Settings-System Settings.	-

Caution :

Please ensure the number of digit on the Assay Parameter Settings - Assay Group Settings is set up

properly. Refer to the caution of [Analysis Parameter, Data] in [5.1. Analysis Data Format] as the detail.

End of Documents